



UNIVERSITÀ DI PARMA



Informatica e Laboratorio di Programmazione

*The true sign of intelligence is not knowledge
but imagination.*

Albert Einstein

- $\sim \frac{2}{3}$ in Class (slides + code examples)
- $\sim \frac{1}{3}$ in Laboratory (actual programming)
- Slides, code examples, and laboratory assignments
 - <https://elly2025.didattica.unipr.it>
 - <https://www.ce.unipr.it/didattica/informatica/>
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- September 18th – December 17th
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- Pauses
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Menù

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[Ritorna](#) alla pagina del corso.

Elenco prove pratiche:

A.A. 2023-2024

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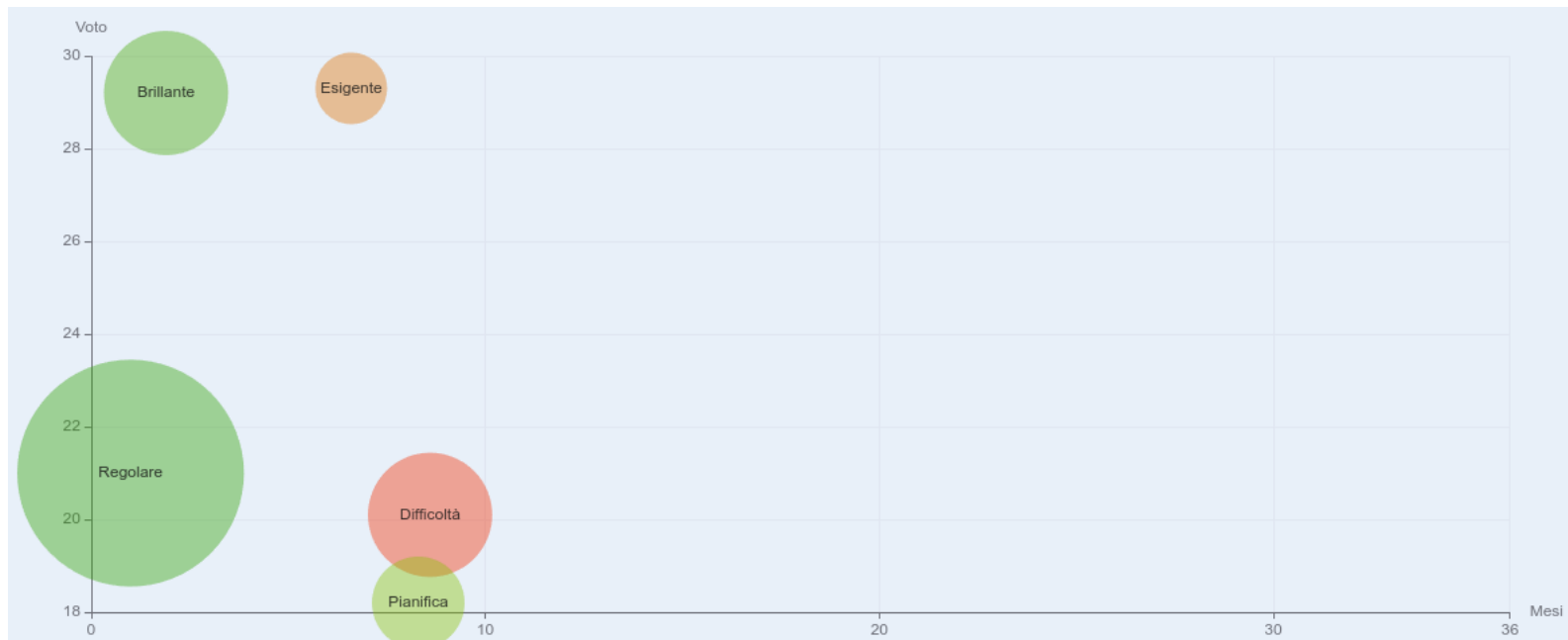
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 - Passed: 52.2% (48 students out of 102)
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 - Time average: 1.9 months
 - Average n. of attempts: 1.8
 - Average result: 24.01

How to approach this course



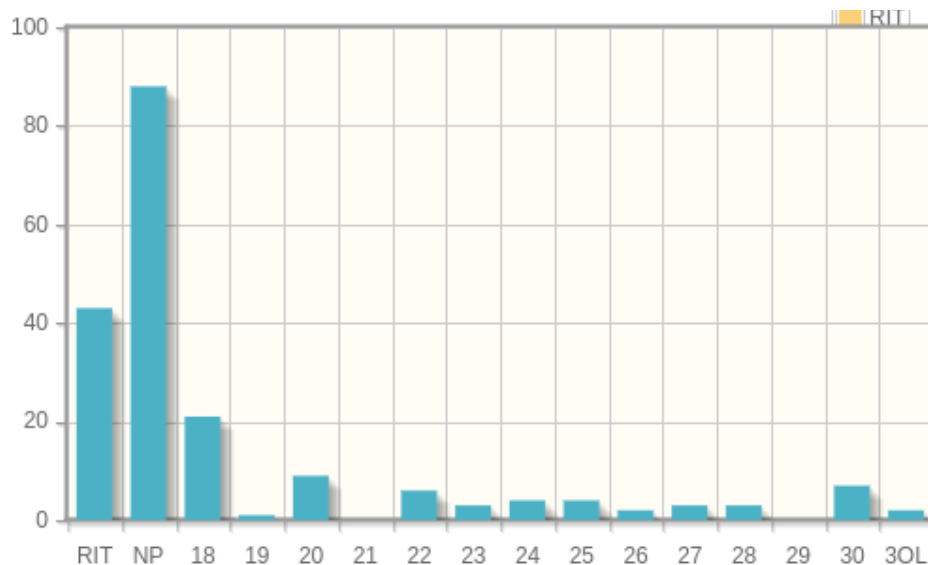
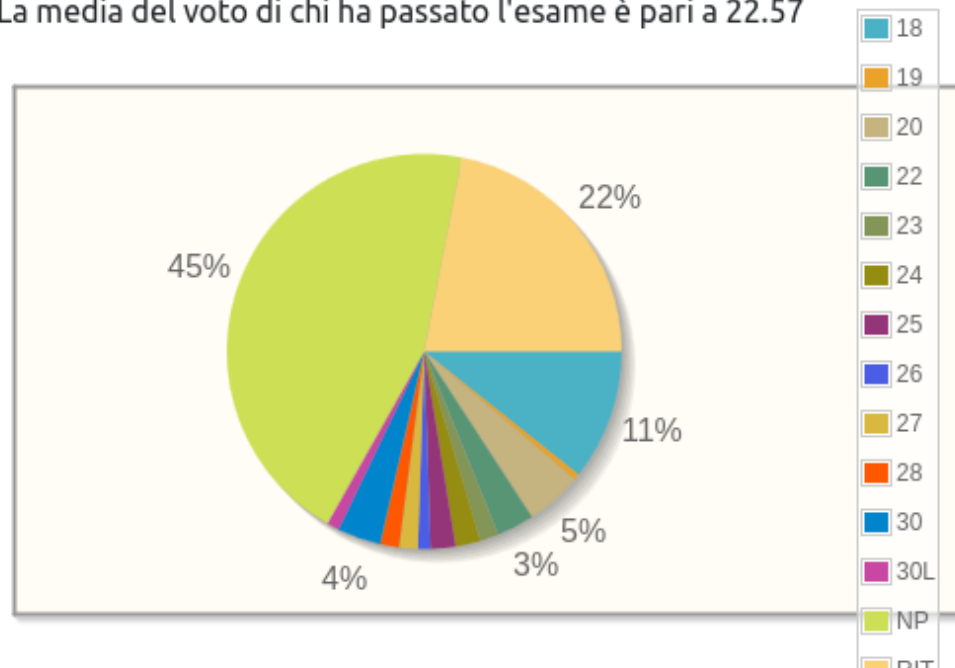
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<i>Returns</i>	c (the character written). If a write error occurs, putchar sets the stream's error indicator and returns EOF.	7.3, 22.4
puts	<i>Write String</i> <code>int puts(const char *s);</code> Writes the string pointed to by s to the stdout stream, then writes a new-line character.	<stdio.h>
<i>Returns</i>	A nonnegative value if successful. Returns EOF if a write error occurs.	13.3, 22.5
putwc	<i>Write Wide Character to File (C99)</i> <code>wint_t putwc(wchar_t c, FILE *stream);</code> Wide-character version of putc.	<wchar.h> 25.5
putwchar	<i>Write Wide Character (C99)</i> <code>wint_t putwchar(wchar_t c);</code> Wide-character version of putchar.	<wchar.h> 25.5
qsort	<i>Sort Array</i> <code>void qsort(void *base, size_t nmemb, size_t size, int (*compar)(const void *, const void *));</code> Sorts the array pointed to by base. The array has nmemb elements, each size bytes long. compar is a pointer to a comparison function. When passed pointers to two array elements, the comparison function must return a negative, zero, or positive integer, depending on whether the first array element is less than, equal to, or greater than the second.	<stdlib.h> 17.7, 26.2
raise	<i>Raise Signal</i> <code>int raise(int sig);</code> Raises the signal whose number is sig.	<signal.h>
<i>Returns</i>	Zero if successful, nonzero otherwise.	24.3
rand	<i>Generate Pseudo-Random Number</i> <code>int rand(void);</code>	<stdlib.h>
<i>Returns</i>	A pseudo-random integer between 0 and RAND_MAX (inclusive).	26.2
realloc	<i>Resize Memory Block</i> <code>void *realloc(void *ptr, size_t size);</code> ptr is assumed to point to a block of memory previously obtained from calloc, malloc, or realloc. realloc allocates a block of size bytes, copying the contents of the old block if necessary.	<stdlib.h>
<i>Returns</i>	A pointer to the beginning of the new memory block. Returns a null pointer if a block of the requested size can't be allocated.	17.3



A.14.2 Pseudo-Random Number Generator Functions

The *rand()* and *srand()* functions enable you to generate pseudo-random numbers.

A.14.2.1 The *rand()* Function

```
#include <stdlib.h>
int rand( void );
```

The *rand()* function returns an integer in the range 0 through *RAND_MAX*. Successive calls to *rand()* should produce different integers. However, the sequence of random numbers could be the same for each program execution unless you use a different seed value via the *srand()* function.

A.14.2.2 The *srand()* Function

```
#include <stdlib.h>
void srand( unsigned int seed );
```

The *srand()* function uses the argument as a seed for a new sequence of pseudo-random numbers to be returned by subsequent calls to *rand()*. If *srand()* is invoked with the same seed value, the sequence of generated numbers will be the same. The default seed value is 1.

A.14.3 Memory Management Functions

The memory management functions enable you to allocate and deallocate memory dynamically. See Chapter 7 for more information about these functions.

A.14.3.1 The *calloc()* Function

```
#include <stdlib.h>
void *calloc( size_t nmemb, size_t size );
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The *calloc()* function allocates contiguous space for *nmemb* objects, each of which has a length in bytes specified by *size*. All bits in the allocated space are initialized to zero. *calloc()* returns a pointer to the first byte of the allocated space. If the space cannot be allocated, or if *nelem* or *size* is zero, *calloc()* returns a null pointer.

- Massimo Bertozzi
- bertozzi@ce.unipr.it
- 0521 90 5845



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KEEP
CALM
IT'S
QUESTION
TIME

nec scire fas est omnia

Quinto Orazio Flacco, Odi

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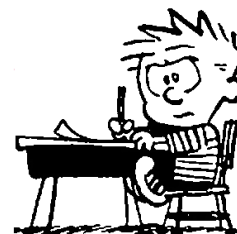
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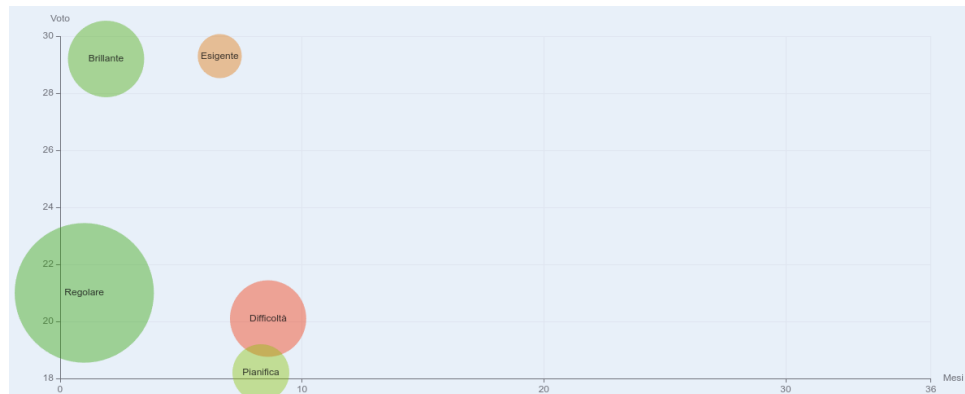
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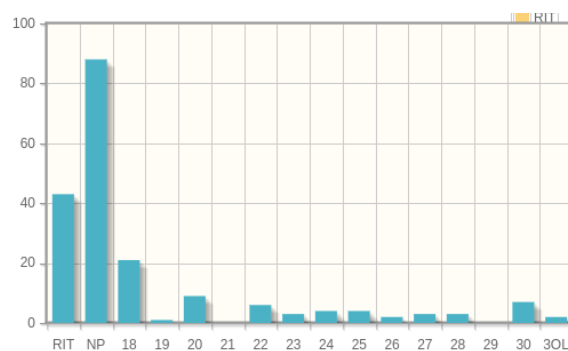
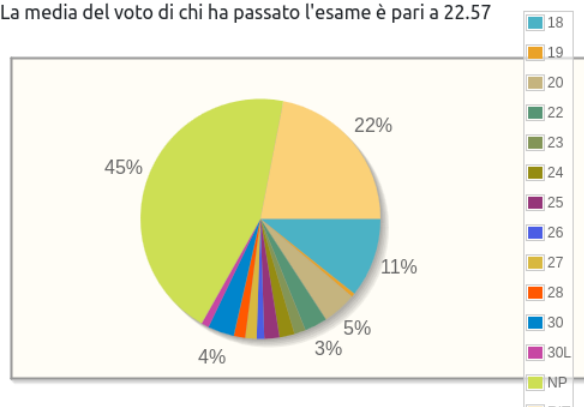
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